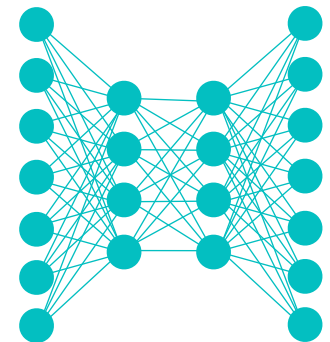


Lecture Notes for **Neural Networks II**



Class Retrospective



Logistics and Agenda

- Logistics
 - Grades!
- Agenda
 - Retrospective!
 - Exam Review



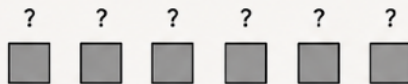
Course Retrospective

NoPE

No Positional Encoding
(Never Took the Course)



- × No signal
- × No context
- × No idea what's happening



All tokens look the same.
No clue. No vibe.

DroPE

Dropped Positional Encoding
(Took It... Then Dropped It)



- × Got the syllabus
- × Saw the first assignment
- × Then vanished into the void



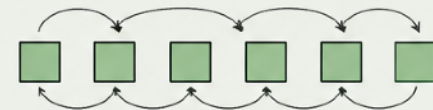
Started with position...
then **dropped** out.

RoPE

Rotary Positional Encoding
(Stuck With It and Succeeded)



- ✓ Built the intuition
- ✓ Survived the assignments
- ✓ Now we're cooking 🍳



Knows where they are,
and how it relates.
Success, in rotation.



Moral of the story: Position matters. Don't be NoPE. Don't become DroPE. Be RoPE.



Course Retrospective

- **Ethics:** The Guidelines, Bias Mitigation, ConceptNet
- **Transformers:** (new) Vanilla, text, vision
- **LLMs:** Encoder, Auto Regressive, SSMs, Emergent, Scaling, GPT
- **CNN Visualization:** Heatmaps, Grad-CAM, Circuits
- **CNN Fully Convolutional:** R-CNN, YOLO, Mask-RCNN, YOLACT, OWL, DETR, Tracking
- **Generation:** VAEs and Stable Diffusion
- Not Covered:
 - **GANs:** Goodfellow to Wasserstein to BigGAN to Stable Diffusion (more)
 - **Style Transfer:** Gatys, FastStyle, WCT
 - **Deep Reinforcement Learning**
 - **Deep Audio Processing**



What score do I need?

- All points in the class add up to 100. Take your existing score and add up to 30 points for the final.
 - Discussion: 10 (ability to present technical topic)
 - Labs: 45 (ability to use topics practically)
 - Exams: 45 (studying forces re-learning, poor assessment)
- Grade Assignment (standard grading for A, B+, B, etc.)
 - 93+ A
 - 90-92.99999 A- Letter of recommendation, yes.

 - 87-89.99999 B+ Letter of recommendation, probably.

 - 83-86.99999 B Letter of recommendation, probably not.
 - 80-82.99999 B-
 - 77-79.99999 C+ 60-69.99999 D
 - 70-76.99999 C Else: F



Final Exam Format

- Final exam is 30% of total class grade
- 30 Multiple Choice Questions
 - 14 covering content from mid-term
 - 11 covering content since mid-term (circuits through stable diffusion)
 - 5 covering papers from semester (future slide)
- 5 Free Response Questions (each question with three parts)
 - 2 covering content from mid-term
 - 3 covering content since mid-term
- MC and FR each worth 50% of exam
- No Extra credit question
- Aiming for 2 hour exam (but you will have three hours)



Final Exam Study Suggestions

- Spend time studying from the **mid-term** exam.
 - 80% time: content you have already been tested over
 - 20% time: content that could have been on the mid-term
- Spend *more time* studying new material
 - **Explainability**: Circuits, Heatmap Visuals, Grad-CAM
 - **Fully convolutional networks**: Architectures and loss functions
 - **Stable diffusion** (high level): flow matching, departure to the latent space, denoising process



Review Papers

- Tiny Recursive Models
- GeoBench
- Fourier Continuous Kernels
- LoRa and Diffusion
- Tracing the thoughts of LLMs
- *Audio analysis papers (maybe not...)*



Example MC Questions

- What is the training complexity of full softmax attention?
- What is the training complexity of linear attention?
- What is the complexity of 1D convolution for a kernel size L , sequence sized T , with d dimensions?
- What is a polysemantic neuron in the context of circuits?
- What fully convolutional architecture uses a region proposal network?



Example MC Questions

- What was the computational bottleneck of RCNN?
- Does ViT have emergent abilities?
- What is $u_t^\theta(x_t)$ in stable diffusion?
- Why does YOLO employ a square root in the bounding box regression of the loss function?
- Why do we use positional encoding in transformers?



On the fly MC?

- What is the key insight in the YOLACT work?
- Suggestions? ...



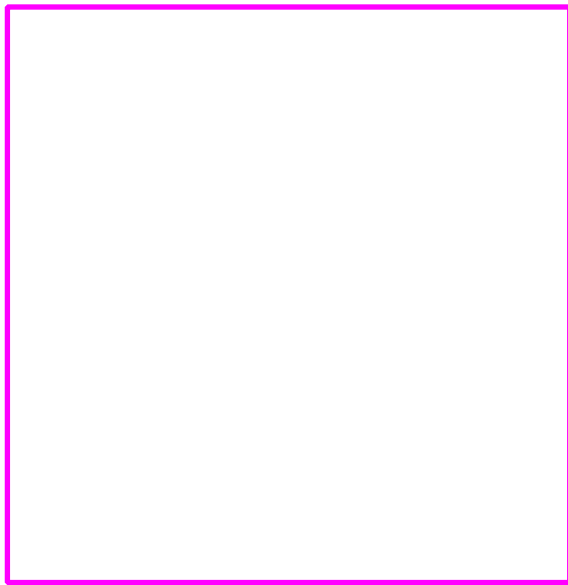
Example FRQ Question

Show the various changes (architecturally) in RCNN as it shifted to (1) Fast-RCNN to (2) FasterRCNN and then to (3) mask-RCNN. At each point, highlight the new portions of the architecture that are most important for speed and performance.



Example FRQ Question

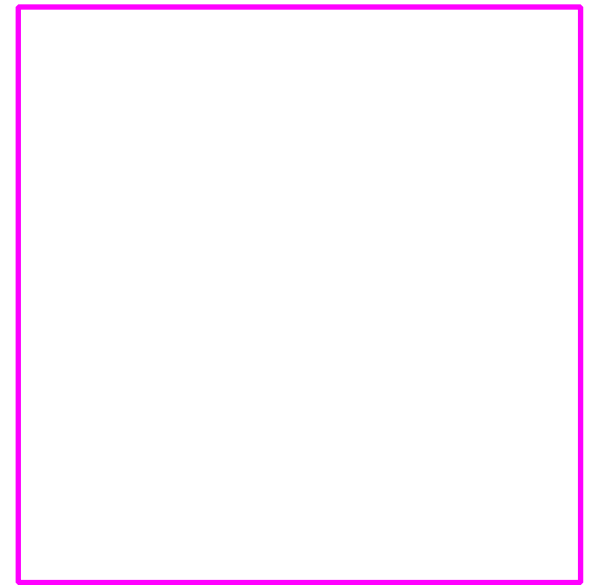
Draw an example vector field used in Stable Diffusion for $d=2$ at three time steps ($t=0$, $t=0.5$, and $t=1$). Initial distribution should be Gaussian and the final distribution should be a single point in 2D space. For each vector field, also draw a few points that might be points on a trajectory for X_0 , $X_{0.5}$, and X_1 .



$t=0$



$t=0.5$

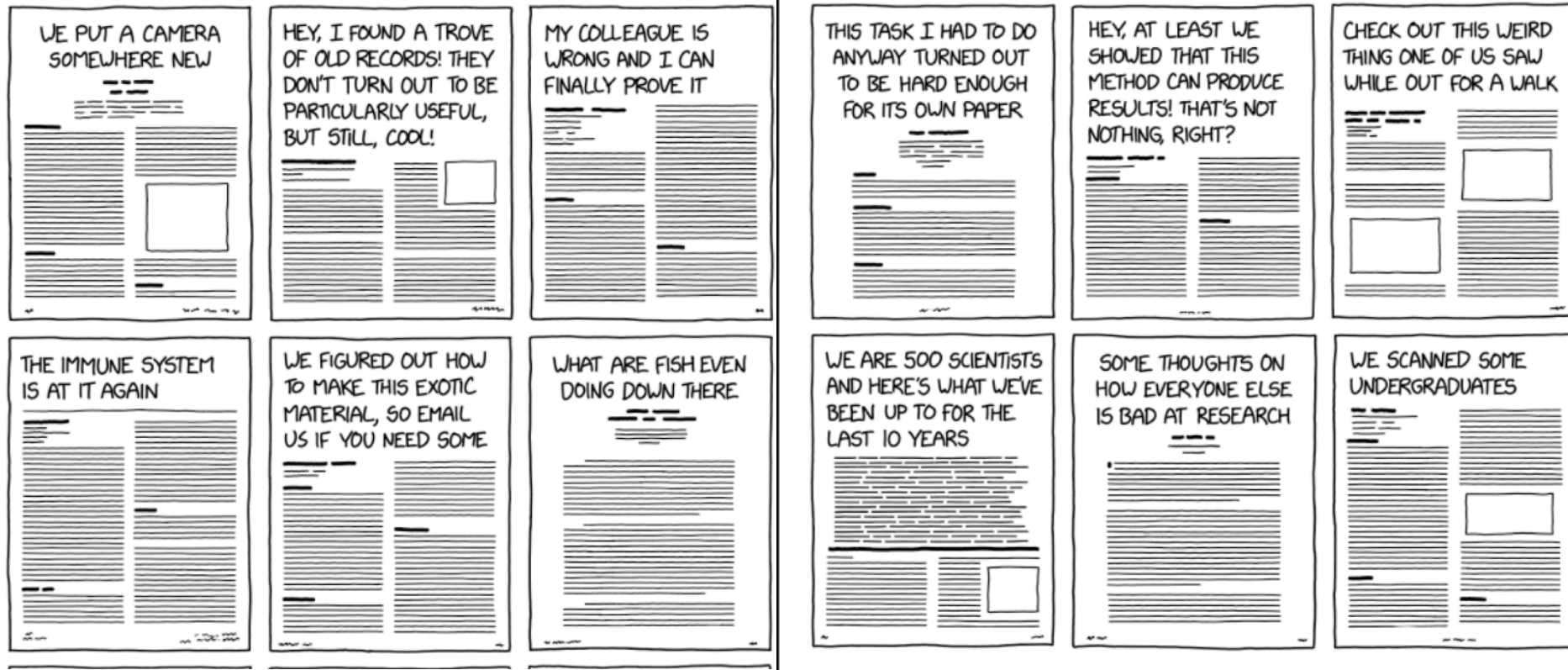


$t=1$

Finally, what would happen at $t=2$?



Types of Scientific Papers



Thanks for a great semester!!!

Please fill out the course evaluations!!



Lecture Notes for **Neural Networks II**

Course Retrospective

Next Time:

None!

Reading: Nope

